

# Alejandro Alfaro

Software Engineer

Bay Area

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## Skills

### Embedded & Systems

**Zephyr RTOS, BLE and IoT Wireless Technology**, Sensor Driver Development, Power Management Electronics, Simulation Technologies for Testing, Tracing/Debugging

### Systems Programming and Scripting

**C++, Python**, Bash, Go

### Mobile Development

**Swift**, Objective-C, Kotlin

### Medical Device Domain Knowledge

**IEC 62304 Development**, Class II Devices, CE Mark and FDA Submissions, IEC 60601

## Experience

### Firmware and Hardware Platform Manager

**Chicago, IL and Remote**

Sibel Health

September 2023 – Present

- Led development of two wearable medical devices — an ICU-grade multi-lead ECG sensor for hospital use and an acousto-mechanical sensor for therapeutics and clinical research — securing over \$10M in grant funding.
- Architected embedded systems for both products from component selection and EVT prototype procurement through early technical de-risking of hardware and firmware designs.
- Implemented low-level drivers for sensing components (digital mics, multi-lead ECG AFE) and critical peripherals (NAND flash, wireless-charging PMICs).
- Designed IPC and thread management on Zephyr RTOS across ARM Cortex-M SoCs spanning single-core (nRF52840), dual-core (nRF5340), and quad-core (nRF54H20) architectures.
- Drove unit test initiative using Native Posix simulation and Antmicro's Renode emulation, growing codebase coverage from 0% to 30% within one month.
- Designed a 3-image bootloader strategy (bootloader + app + firmware loader) that doubled available ROM from 400KB to 800KB on memory-constrained platforms, added OTA recovery mode, and isolated BLE image downloads from the application for improved security.

### Technical Project Manager

**Chicago, IL and Remote**

Sibel Health

September 2023 – Present

- Managed the Design History File (DHF) for two next-gen platforms through Phase 1-3 of a Medical Device PDP, collaborating with multi-disciplinary teams to finalize design inputs before production.
- Reduced engineering costs by ~\$1-2M by outsourcing hardware and firmware development to an offshore partner with strong silicon vendor relationships, accelerating component procurement.
- Coordinated between the offshore team and Sibel leadership through design reviews and regular syncs, ensuring Sibel retained ownership of key technical decisions while meeting design requirements.

### Audio & Haptics Software Engineer

**Cupertino, CA**

Apple

March 2022 – September 2023

- Optimized Apple's CoreHaptics framework, achieving 20% less CPU usage across haptic and audio playback scenarios.
- Fixed a fundamental sound wave interpolation bug in Apple's haptic synthesizer, eliminating Taptic Engine motion glitches during sustained haptic playback.
- Improved iOS 16 haptic keyboard quality by validating behavior across all audio routes (wireless/wired headphones, CarPlay, AirPlay, HDMI) and expanded CI/CD test coverage for the CoreHaptics framework.

## Firmware Engineer

Niles, IL

Prior Experience at Sibel Health

June 2020 – March 2022

- Led firmware development for the Maxim Integrated Health Patch Platform MAXREFDES282 , a reference design showcasing the MAX816178 PPG, ECG, and BIOZ all-in-one AFE
- Built a fleet management system to verify quality of medical sensors manufactured abroad, reducing defective shipments by 21%. Developed diagnostic firmware routines for PCB integrity validation and life-cycle stress testing.

## Education

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### Northwestern University

Evanston, IL

Bachelor's and Master's in Mechanical Engineering

Aug 2015 – Dec 2020

- Relevant Coursework: Mechatronics, Data Structures & Algorithms, Feedback Control Systems, Embedded Systems in Robotics, SLAM

## Interests

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Surfing, The Sopranos, long distance running, and electronic music production